Tech Bite #4: Electricity Basics -- Voltage, Current, and Power

The three words every radio runs on -- voltage, current, and power -- plus conductors, insulators, and the difference between AC and DC.

Group T5A · 11 pool questions · 5-10 minute presentation · Practice exams at eham.net/exams

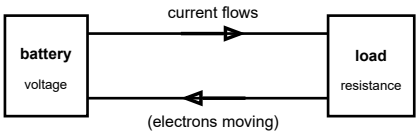
The Big Picture

Before any radio makes sense, you need three words: **voltage**, **current**, and **power**. This Bite is short and plain – it is the foundation the math in Bite 5 is built on. Nothing here is hard; it is just vocabulary, and getting the vocabulary right makes everything after it easier.

Voltage, Current, and Power

Electricity is moving electrons. The **flow of electrons** through a circuit is called **current**, and current is measured in **amperes** (amps). (T5A01, T5A03)

What makes the electrons move is a difference in **voltage** – voltage is the "push" behind the current. (T5A05) And **power** – the rate at which electrical energy is used – is measured in **watts**. (T5A02, T5A10)



Voltage is the push; current is the electron flow it drives; the load uses the power.

Conductors and Insulators

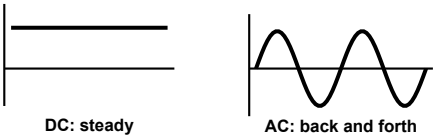
Some materials let current flow easily; others block it. **Metals** are good **conductors** because they have **many free electrons** available to move. (T5A07) **Glass** is a good **insulator** – it has almost no free electrons, so current cannot pass through it. (T5A08)

AC and DC

Current comes in two kinds. **Direct current (DC)** – like the current from a battery – flows steadily in **one direction**. **Alternating current (AC)** – like household power – **alternates between positive and negative directions**, back and forth. (T5A09)

The number of complete back-and-forth cycles each second is the **frequency**, and frequency is measured in **hertz**. (T5A04, T5A06)

Resistance opposes current – all of it: DC, AC, and RF alike. (T5A11)



DC flows steadily one way; AC reverses direction over and over. Cycles per second is the frequency.

Mental Cheat Sheet

- **Current** = flow of electrons; measured in **amperes**.
- **Voltage** = the push that drives current; a **voltage difference** causes electron flow.
- **Power** = rate of using electrical energy; measured in **watts**.
- **Frequency** = cycles per second; measured in **hertz**.
- Metals conduct because they have **many free electrons**.
- **Glass** is a good **insulator**.
- **DC** flows one direction; **AC** reverses back and forth.
- **Resistance** opposes current – DC, AC, and RF.

Common Traps

- Current is in **amperes**, power is in **watts** – do not swap them.
- Watt-**hours** measure energy used over time, not power itself.
- It is a **voltage** difference that causes electron flow – not capacitance or inductance.
- Metals conduct because of free **electrons**, not free protons.
- AC alternates between **positive and negative** – not just "positive and zero."
- Resistance opposes **all** current types, not only DC.

Takeaway: Electricity is moving electrons. **Voltage** is the push, **current** (amperes) is the flow it causes, and **power** (watts) is the rate energy is used. Frequency, measured in **hertz**, counts how many times per second AC reverses. Conductors like metal carry current easily; insulators like glass block it. Learn these words now – Bite 5 turns them into simple math.

Bite 4 Question Bank (T5A)

T5A01. Electrical current is measured in which of the following units?
A. Volts
B. Watts
C. Ohms
D. Amperes

T5A02. Electrical power is measured in which of the following units?
A. Volts
B. Watts
C. Watt-hours
D. Amperes

T5A03. What is the term for the flow of electrons in an electric circuit?
A. Voltage
B. Amperes per second
C. Volts per second
D. Current

T5A04. What term describes the number of times per second that an alternating current makes a complete cycle?
A. Pulse rate
B. Wave number
C. Frequency
D. Wavelength

T5A05. A difference in which of the following causes electron flow?
A. Voltage
B. Ampere-hours
C. Capacitance
D. Inductance

T5A06. What is the unit of frequency?
A. Hertz
B. Henry
C. Farad
D. Epicycles per second

T5A07. Why are metals generally good conductors of electricity?
A. They have relatively high density
B. They have many free electrons
C. They have many free protons
D. They have relatively high Young's modulus values

T5A08. Which of the following is a good electrical insulator?
A. Sea water
B. Glass
C. Stainless steel
D. Graphite

T5A09. Which of the following describes alternating current?
A. Current that alternates between a positive direction and zero
B. Current that alternates between a negative direction and zero
C. Current that alternates between positive and negative directions
D. All these answers are correct

T5A10. Which term describes the rate at which electrical energy is used?
A. Resistance
B. Current
C. Power
D. Voltage

T5A11. What type of current flow is opposed by resistance?
A. Direct current
B. Alternating current
C. RF current
D. All these choices are correct

Answer Key

T5A01: D T5A02: B T5A03: D T5A04: C

T5A05: A T5A06: A T5A07: B T5A08: B

T5A09: C T5A10: C T5A11: D